Palaeomagnetism and chronology of Caledonian igneous episodes in the Cross Fell Inlier and northern Lake District

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This study employs palaeomagnetism to differentiate between Palaeozoic minor intrusions in northern England. The lamprophyres of the Cross Fell Inlier subdivide petrographically into kersantites and minettes. All the kersantites have steep directions of magnetisation probably correlating with an interval of anomalous geomagnetic field in Caradocian times. The Threlkeld–St John's microgranodiorite (445 m.y.) and Carrock Fell dolerite dyke swarm also possess steep magnetisations assigned to the same interval. The minettes have shallow directions of magnetisation similar to the Shap Granite (393 m.y.) and correlate with the later Caledonian intrusions of northern England. These episodes of intrusion are separated by late-Silurian tectonism and define a trend of increasing K_2O/Na_2O in the later magmas of the Caledonian orogeny. Miscellaneous palaeomagnetic results from the Borrowdale Volcanic Group, Embleton diorites and Haweswater dolerite are also reported.

1. Introduction

In certain circumstances the palaeomagnetic method can be employed to date episodes of igneous intrusion where reliable age data cannot otherwise be obtained by conventional radiometric methods. It involves isolating primary directions of magnetisation from the rock body, determining the corresponding palaeomagnetic pole, and comparing this pole with the apparent polar wander (a.p.w.) path for the region independently established from studies on dated rocks. The palaeomagnetic method of dating has the advantage that many more rock units can be surveyed than is practicable with the Rb–Sr method, and it does not suffer from the difficulties of interpretation caused by variable argon loss in the K–Ar method. However, a successful application depends critically on a number of factors. Firstly, the age of the rock unit must be known within broad limits; secondly the a.p.w. path must be well defined between these limits, and thirdly, a.p.w. movements should be both relatively rapid and occur frequently for the method to be of much use. The difficulties of applying palaeomagnetism to dating rocks are acute in Precambrian times when the palaeopole is known to have undergone excursions and returned to similar positions, sometimes within periods of 100 m.y., while at other times it remained essentially stationary for periods of tens of millions of years.

The a.p.w. path for Britain in Lower Palaeozoic times is now sufficiently well understood for it to be considered as a practical dating tool for this interval (Briden

et al. 1973; Faller *et al.* 1976; Piper 1978). Studies of many dated rocks ranging from Aberdeenshire to Somerset show that the pole lay in the vicinity of 10°E, 10°S during much of Ordovician times, and near 320°E, 0°N from Lower Wenlock to the end of Lower Devonian times (*c.* 430–390 m.y.). Some rocks assigned to the late Llandeilo–early Caradocian and Ashgill to Llandovery intervals possess anomalous magnetic directions yielding palaeomagnetic poles quite different from those quoted above (Thomas and Briden 1976; Piper 1978). Whether these anomalous poles represent long-term geomagnetic field anomalies or rapid plate movements is difficult to decide from the present data, but is not relevant to the present work which aims to make use of these differing palaeofield directions.

This study commenced as an inquiry firstly into the possibilities of distinguishing between minor intrusions on the basis of their palaeomagnetism, and secondly to determine whether this would permit them to be related to major intrusive bodies for which radiometric data already existed. This has necessitated extending the palaeomagnetic examination to the major intrusions. In the event it has proved possible to assign ages to all of the intrusive episodes studied within fairly narrow limits, and to place them in their context of the regional tectonic and magmatic development during the latter part of the Caledonian cycle.

2. Geological outline

2a. Lake District igneous rocks

Deposition of the rocks now comprising the Skiddaw Slate Group of northern England was succeeded by a whole range of intrusive and extrusive igneous rocks which may, in part, be genetically related to intrusion of the Caledonian batholith underlying the Lake District Dome and defined by gravity study (Bott 1974). The research of recent years has, however, shown that these igneous episodes occurred at several different periods between and after Caledonian tectonic events. For example, eruption of the Eycott Group of the north Lake District comprising basalts and basaltic andesites (Eastwood *et al.* 1965) commenced during the Didymograptus *bifidus* time zone of the Llanvirn (Downie and Soper 1972), and was tilted prior to eruption of the Borrowdale Volcanic Group (palaeomagnetic sites 1 to 4 of this study), comprising predominantly andesites and rhyolites (Firman 1957; Mitchell 1956; Moseley 1960, 1964). This latter group lies unconformably on Llanvirnian slates east of Keswick but may be essentially conformable in the southwest of the Lake District outcrop; it was in turn folded and partly eroded prior to deposition of the (Caradocian) Coniston Limestone Group including minor volcanic components (Eastwood 1953).

The Haweswater dolerite (Fig. 1 and palaeomagnetic sites 5–7 of this study) intrudes the Borrowdale Volcanic Group on either side of Haweswater and was interpreted as a vent related to the group by Nutt (1968). Further to the north the Threlkeld–St John's microgranodiorite (sites 8–14 of this study, Figs 1 and 2) intrudes the base of the Borrowdale Volcanic Group; the study of Wadge *et al.* (1976) confirms that this body is Ordovician in age (Rb–Sr isochron, 445±15 m.y.). In the vicinity of Embleton, northwest of Bassenthwaite, the Skiddaw Slates are intruded by a suite of many minor intrusions (palaeomagnetic sites 22–25 of this study, Fig. 2B), and a swarm of W–NW trending dolerite dykes cuts members of the Eycott Group in the vicinity of Carrock Fell (palaeomagnetic sites 15 to 21, Fig. 2A).

A distinctly later episode of Devonian igneous intrusion postdates the regional end-Silurian folding and includes the Shap granite dated 393 m.y. (mean of K–Ar and Rb–Sr whole rock ages, Brown *et al.* 1964); samples were also collected from this unit

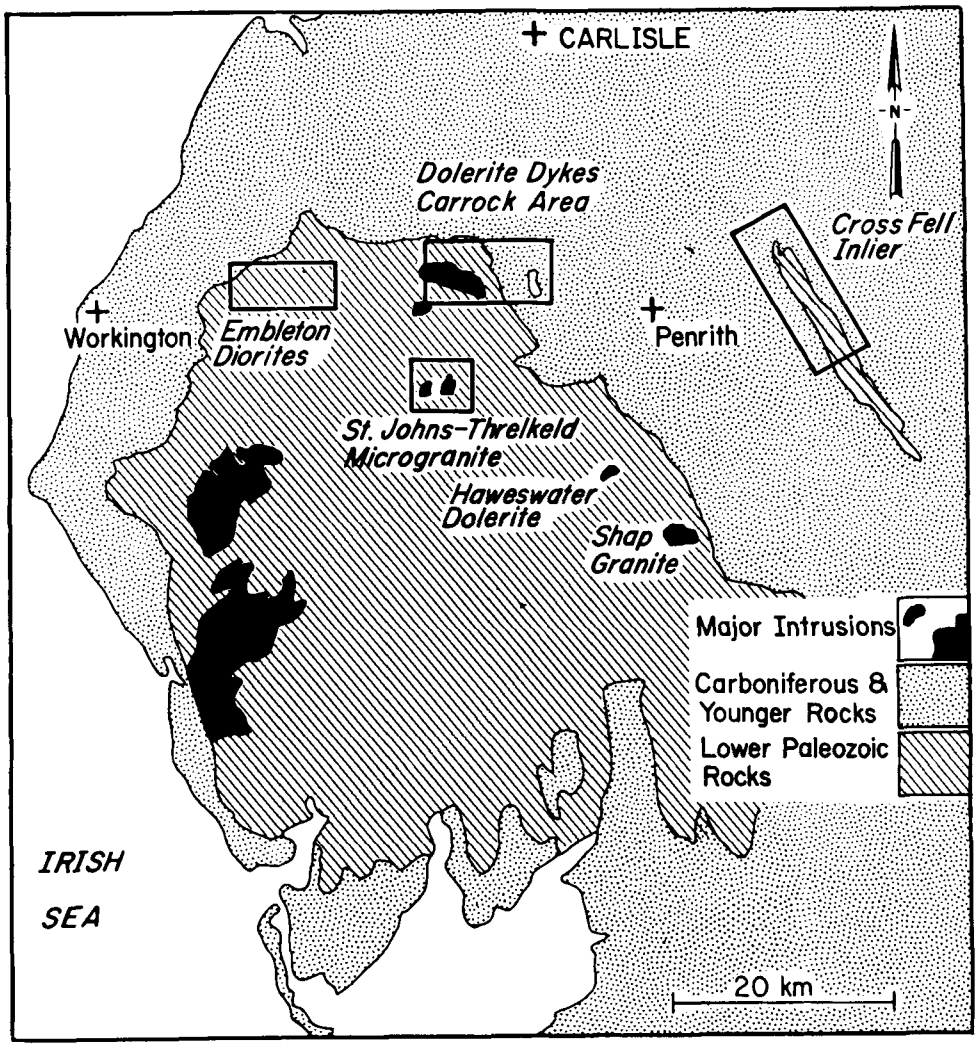


Fig. 1. The Lake District and Cross Fell Lower Palaeozoic inliers showing the locations of rock units sampled for this study.

for palaeomagnetic study. Gravity studies by Bott (1967 and 1974) have shown a ridge in the Caledonian batholith extending from the Alston block across to the Shap outcrop of the sub-Lake District batholith. Samples of the former taken from the Rookhope borehole on Weardale were dated by Holland and Lambert (1970) using a Rb–Sr whole rock isochron; the age of 394 ± 34 m.y. (see Dunham 1974) adds a temporal relationship to that deduced spatially by gravity surveys between the Skiddaw (398 m.y.), Shap and Weardale granites.

2b. Cross Fell intrusions

In the Cross Fell Inlier (Fig. 3) folded Ordovician and Silurian lavas and sediments are cut by many minor intrusions which are most abundant in the northern part of the inlier. Petrologically the minor intrusions are predominantly felsites with a E–NE

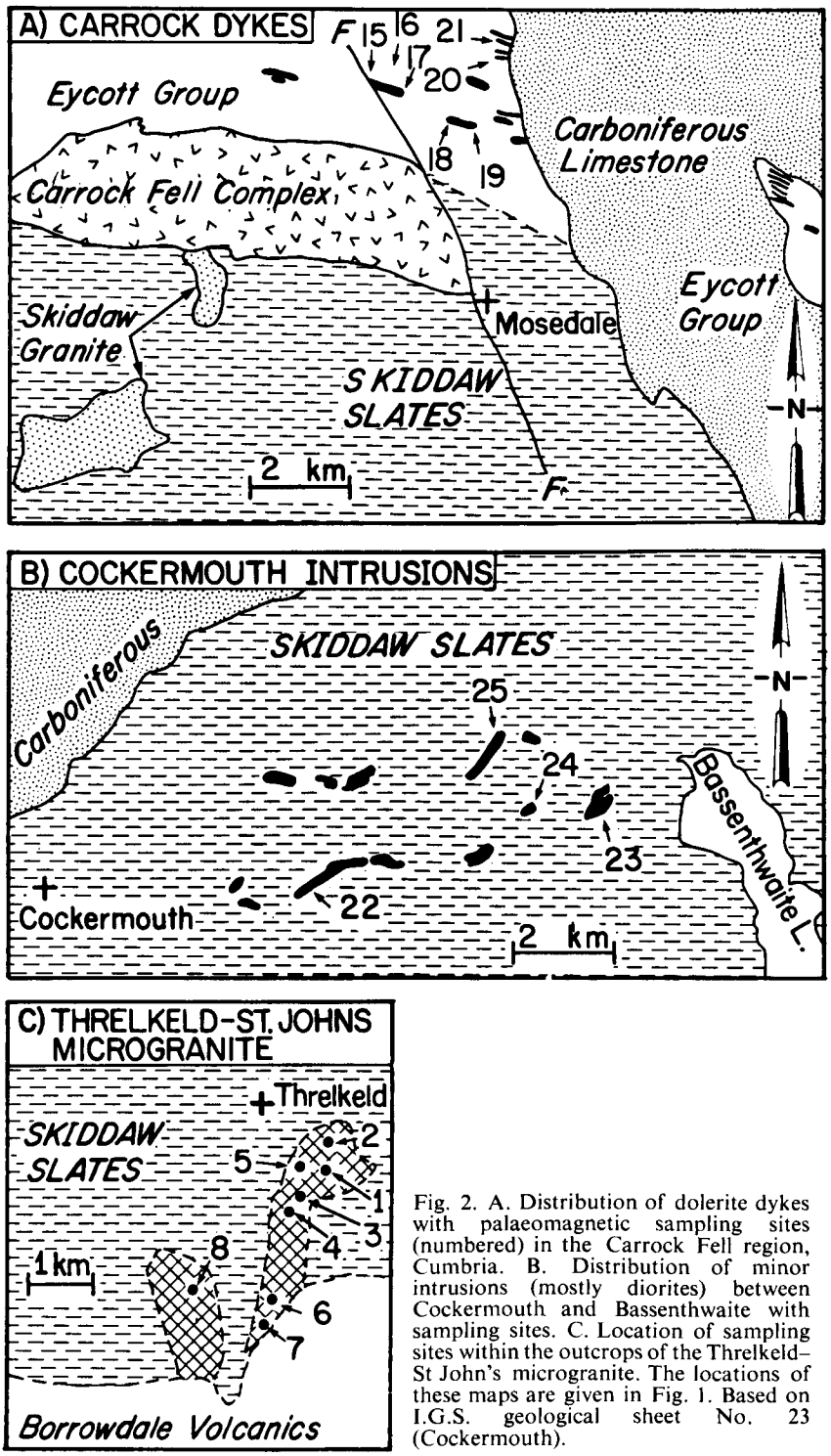


Fig. 2. A. Distribution of dolerite dykes with palaeomagnetic sampling sites (numbered) in the Carrock Fell region, Cumbria. B. Distribution of minor intrusions (mostly diorites) between Cockermouth and Bassenthwaite with sampling sites. C. Location of sampling sites within the outcrops of the Threlkeld–St John's microgranite. The locations of these maps are given in Fig. 1. Based on I.G.S. geological sheet No. 23 (Cockermouth).

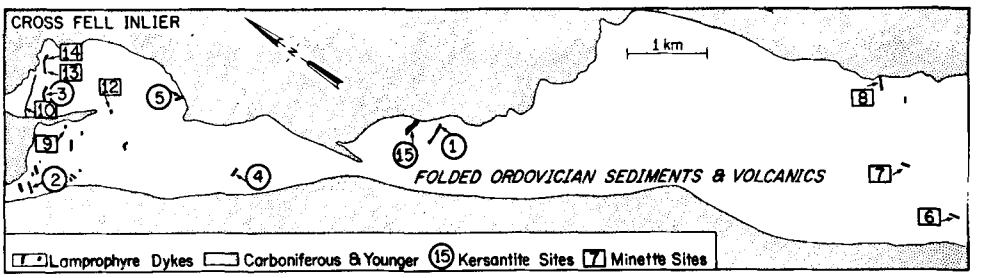


Fig. 3. Distribution of minor intrusions in the northern part of the Cross Fell inlier with palaeomagnetic sampling sites. Based on geological special sheet: Cross Fell Inlier.

trend parallel to the regional Caledonian structures, and lamprophyres. For technical reasons this study has been concentrated on the latter (Fig. 3). The lamprophyres are poorly exposed, often highly weathered, grey or brown rocks which show such slight baked margin effects that they must have been intruded at low magmatic temperatures (Burgess and Wadge 1974). They cannot readily be differentiated in the field and their petrologic affinities have only been resolved by thorough analysis of thin sections. On this basis the dykes can be defined either as kersantites (sites 1 to 5 and a site 15 (Fig. 3) which did not yield significant palaeomagnetic results) or minettes (sites 6 to 14). Modal analyses of kersantite dykes yield the following estimates of mineralogical composition:

	Site 1	*Site 4*
Plagioclase	49%	58%
Biotite	28%	28%
Chloritised augite	9%	
Quartz	5%	8%
Calcite	6%	4%
Apatite, reibeckite, iron ores	3%	4%

Modal analyses for two minettes yield the following estimates of composition:

	Site 6	*Site 12*
Biotite	32%	40%
Orthoclase	30%	50%
Diopside	18%	5%
Ti-augite	7%	—
Calcite	8%	5%
Apatite, sphene, iron ores	4%	
Quartz xenocrysts	1%	—

It is probable that the extensive lamprophyric-type hydrothermal alteration took place at relatively high deuteric temperatures (*c.* 500–800°C) as an autometamorphic effect because the biotite crystals have reaction rims and titanomagnetites are the principal magnetic remanence carriers. However, the mineralogical evidence is not diagnostic because kaolinisation of feldspars is known to occur at lower temperatures of 400–100°C. Both demagnetisation characteristics (Fig. 5) and Curie point determinations (yielding single Curie points between $T_c = 480$°C and 570°C) confirm the presence of titanomagnetite; high coercive force components and Curie points above 570°C indicative of hematite have not been identified in any of the sample rock units.

The minettes and kersantites also appear to be structurally distinct, and all the kersantites except site 2 are dykes with an approximately east-west strike (Fig 3). It cannot be established in the field that all the minettes are in fact dykes, but where they have a well defined strike it is approximately NE–SW or towards the outcrop of the Shap Granite. The lamprophyres all cut folded Arenig and Llanvirn sediments with the exception of minette site 6 which cuts the Silurian (Llandovery) Browgill Beds.

Hudson (1937) took the view that the close geographical link between the lamprophyres and felsites of Cross Fell suggested these two types were differentiates of the same parental magma (see also Burgess and Wadge 1974). While not disputing this probability, the present study was envisaged as a means of testing Hudson's other suggestion of a close temporal link between the parental magmas and those of certain Lake District intrusions (see section 4). The extent of any time difference between the petrographically and structurally distinct lamprophyre suites also might be evaluated.

3. Palaeomagnetic results

3a. Method

Samples were drilled in the field using a portable motorised drill and oriented while still attached to the outcrop by sights on the sun or by magnetic bearings in conjunction with topographic sightings. A 2·5 cm long cylinder was cut from the base of each core and measured before and after progressive demagnetisation in alternating fields; the demagnetising treatment was normally continued in steps of 50 or 100 Oersteds (oe.) for all samples until either no further significant change in direction was observed or until the treatment began to introduce unstable components of magnetisation. All measurements were made with a parastatic magnetometer except those from six sites (16 to 21) in dykes intruding the Eycott Group, which were measured with a PAR SM–2D spinner magnetometer. Portions of the field cores were utilised for study of opaque phases in reflected light, and Curie Point determinations using a Chevallier balance with a strong permanent magnet (~2000 oe.) to saturate the samples; this is an aid to the identification of magnetic phases although the system used is not fully capable of saturating hematite which may have escaped detection in some instances.

3b. Borrowdale Volcanics and Haweswater Dolerite

Incidental to the main study, four sites were sampled from andesitic tuffs of the Borrowdale Volcanic Series from the southeast side of Haweswater and included in the Haweswater Dam andesite and tuff division of Nutt (1966). With magnetic remanent intensities in the range 1×10^{-3} to 1×10^{-5} e.m.u./gm the samples all carried high coercivity remanence (Curie point, Tc ~520°C) very stable to a.f. treatment, and yield site means listed in Table 1. A previous palaeomagnetic survey of Borrowdale Volcanics from the Haweswater–Ullswater area (Morris 1973; Faller *et al.* 1976) yielded 14 sites with dip-corrected directions of magnetisation similar to this group of four. The combined palaeomagnetic pole for all 18 sites is listed in Table 1.

Samples were collected at four sites in the Haweswater dolerite along the northwest shore of Haweswater; this is an ophitic dolerite with albite/oligoclase variably altered to sericite and pyroxene to amphibole. It is a highly alkaline basic unit ($\Sigma Na_2O + K_2O \approx 5{\cdot}5\%$) and has frequently been designated a camptonitic lamprophyre, closely related to kersantite on the basis of its abundant sodic plagioclase. The dolerite contains about 2% opaque minerals and Curie point determinations of approximately 550°C confirm that titanomagnetite is the main remanence carrier. Polished sections exhibit subhedral to euhedral unoxidised titanomagnetite grains (30–200μ) with slight

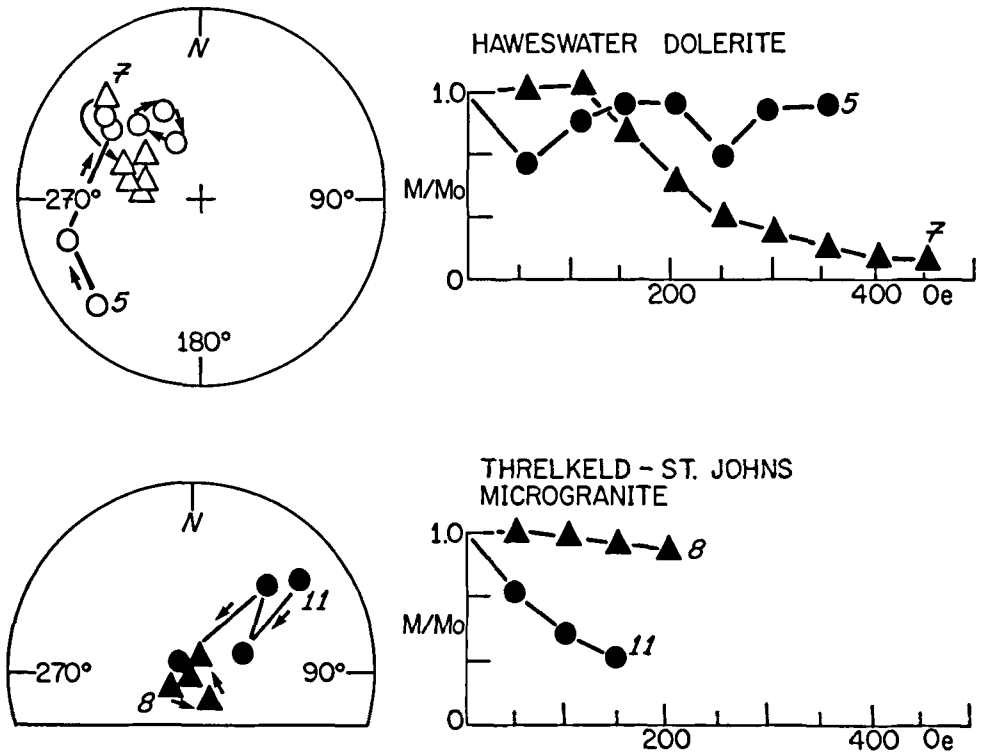


Fig. 4. Behaviour of directions of magnetisation and magnetic moments with progressive a.f. demagnetisation of some typical samples from the Haweswater Dolerite, and Threlkeld-St John's Microgranite The projections are polar stereographic where the open symbols are upper and closed symbols are lower hemisphere plots. The behaviour of magnetic moments is expressed as a fraction of the moment Mo prior to a.f. treatment. Numbers refer to sites in the Haweswater Dolerite and Threlkeld–St John's microgranite Symbols on graphs and plots refer to the same samples.

marginal maghemitisation, and agglomerations of sulphide grains (2–10μ). Around some titanomagnetite grains the marginal maghemite has been further altered to hematite and entire grain areas are speckled with secondary rutile; this is probably a granulation effect (Ade-Hall *et al.* 1971) resulting from later limited hydrothermal metamorphism.

Three of these sites with NRM intensities of the order of 1×10^{-5} e.m.u./gm stabilise after demagnetisation in low fields (50–100 oe.) with little or no movement of magnetic vectors in higher fields (Fig. 4); the fourth site exhibited weaker intensities, was less stable, and is excluded from further consideration. Morris (1973) and Faller *et al.* (1976) report palaeomagnetic data from one site (73E) in the Haweswater dolerite and three sites from the aureole (69E,70E and 71E) in the Borrowdale Volcanic Group. The site 73E has a similar inclination to the sites of this study but an anomalous declination (249°) and the palaeomagnetic pole position of Table 1 is accordingly calculated using only the sites of this study.

3c. Embleton Diorites

This term is used here to embrace a suite of minor diorite and lamprophyre intrusions

Table 1. Palaeomagnetic Results, Palaeozoic igneous rocks of the Lake District

	N	D	I	R	k	α_{95}	
1. Borrowdale Volcanics, Haweswater (tilt corrected)							
1	6	341	−45	5·89	47	10	
2	6	346	−43	5·97	219	5	
3	6	324	−64	5·99	612	3	*Palaeomagnetic Pole* °E °N 198·8 6·2
4	5	314	−44	4·90	41	12	
Mean, 18 sites*		335·8	−45·1	17·22	22	7·6	($d\Psi = 6{\cdot}1$, $d\chi = 9{\cdot}6$)
2. Haweswater Dolerite							
5	6	310	−47	6·67	18	15	
6	5	328	−59	4·76	19	18	*Palaeomagnetic Pole* °E °N
7	6	300	−69	5·72	18	16	212·1 −11·8
Mean, 3 sites		314·7	−59·5	2·94	36	21·8	($d\Psi = 22{\cdot}9$, $d\chi = 30{\cdot}9$)
3. Threlkeld–St Johns Microgranodiorite							
8	5	248	88	4·39	7	32	
9	4	36	41	3·85	20	21	
10	7	32	56	6·48	12	18	
11	4	31	75	3·72	11	29	
12	3	57	63	2·98	89	13	
13†	4	163	31	3·95	63	12	*Palaeomagnetic Pole* °E °N
14	8	11	66	7·47	13	16	88·0 68·3
Mean, 6 sites		34·2	66·4	5·76	21	15·0	($d\Psi = 20{\cdot}3$, $d\chi = 247$)

4. E–W Dolerite dykes intruding Eycott Group‡							
15	7	9	54	6·94	103	7	
16	6	6	45	5·89	47	10	
17	7	2	45	6·95	120	6	
18	6	35	64	5·87	39	10	
19	4	75	63	3·68	9	32	
20	6	29	57	5·99	327	4	*Palaeomagnetic Pole* °E °N
21	6	32	62	5·99	497	3	124·3 67·7
Mean, 7 sites		22·6	57·7	6·78	27	11·7	($d\Psi = 12·6°$, $d\chi = 17·2°$)
5. Embleton diorites, Cockermouth District							
22	6	152	65	5·96	134	6	
23	6	281	−3	5·98	327	4	
24	6	25	−31	5·74	19	15	(VGP: 16°E, 14°N)
6. Shap Granite							
Mean, 12 samples		202·9	−8·2	9·85	5	21·3	(VGP: 328°E, 36°S)

Footnote: D and I are the mean declination and inclination of the remanence vector derived from N sites or samples with a resultant vector R and precision k defined by $k = (N-1)/(N-R)$. α_{95} is the radius in degrees of the 95 per cent confidence circle about the mean direction and $d\Psi$ and $d\chi$ are the semi-axes of the 95 per cent confidence oval about the mean pole in the co-latitude direction and at right angles to it respectively.

*Combined with 14 sites from the Eastern Borrowdale Volcanics given by Faller *et al.* (1976), VGP = Virtual Geomagnetic Pole.

†Sites excluded from calculation of the mean.

‡Corrected for dip of overlying Carboniferous sediments.

in the Cockermouth region northeast of Bassenthwaite Lake; they have a poorly-defined average WSW–ENE strike and intrude Skiddaw Slates. Samples from sites in members of this suite (Fig. 2B) were highly stable to a.f. demagnetisation (the fourth site was too weak to measure) and yield closely-grouped site mean directions (Table 1). In spite of the high quality of this palaeomagnetic information the mean directions do not agree with one another. It is possible that only site 22 is representative of the age of these intrusions because it yields a virtual geomagnetic pole close to other Ordovician results, and has remanent intensities ($0{\cdot}70$–$0{\cdot}30 \times 10^{-5}$ e.m.u./gm) up to an order higher than the petrographically-similar sites 23 and 24 ($0{\cdot}33$–$0{\cdot}05 \times 10^{-5}$ e.m.u./gm; these latter sites may then have become magnetised during a polarity transition when weak fields with intermediate directions pertained. More sites would be needed to resolve this point with certainty.

3d. Threlkeld–St Johns Microgranodiorite

This is a grey or pink medium to fine grained granodiorite of considerably lower K_2O/Na_2O ratio than the Shap granite (Table 3) and consequently less prominent biotite and orthoclase phenocrysts but more oligoclase in an equigranular matrix. It is exposed in three separated quarry outcrops (Fig. 2C) and intrudes the Skiddaaw Slates and lower units of the overlying Borrowdale Volcanic Series (Wadge *et al.* 1976). With NRM intensities of about 10×10^{-6} e.m.u./gm, some samples became too weak to measure after demagnetising in low fields of about 250 oe. However, they remain relatively stable up to this field (Fig. 4) and group to yield mean directions removed from the present geomagnetic-field in this area (Table 1). Petrographic examination confirms the paucity of possible remanence carriers; the most abundant opaque constituents are pseudomorphed grains ($\sim 10\mu$) apparently comprising aggregates of hematite and rutile, but Curie point determinations (Tc = 540°C, Fig. 5) suggest that the sparse grains of titanomagnetite (1–20μ) are the main carriers. The mean palaeomagnetic pole in Table 1 is calculated from six of the seven sites yielding directions significant at the 95 per cent. confidence level.

3e. Dolerite dykes intruding the Eycott Group

Seven sites in the Carrock Fell area from four separate dolerite dykes at Carrock Beck, Linewath Farm and Hegle End (Fig. 2A) yield significant directions of magnetisation before and after (Table 1) a.f. treatment; demagnetisation behaviour and Curie points (Tc ~ 549°C) confirm that titanomagnetite is the remanence carrier. The site-mean directions have been corrected for tilt of the adjacent Carboniferous rocks (which they do not cut), but not for the dip of the (Llanvirn) Eycott Group which they postdate. The group-mean direction of magnetisation is quite different from that of the nearby (pre-Caradocian) Carrock Fell complex (Briden and Morris 1973; Morris 1973, Faller *et al.* 1976) suggesting that the dykes and Carrock complex are not genetically-linked contrary to the evidence of their comparable strike and petrologic similarity (Eastwood *et al.* 1965).

3f. Shap Granite

The so-called Shap "Granite" is a coarse pink-coloured porphyritic adamellite ($K_2O/Na_2O \approx 1{\cdot}38$, Table 3) with large orthoclase phenocrysts set in an equigranular biotite, quartz, feldspar groundmass. Samples (remanent intensities 8×10^{-6}–1×10^{-7} e.m.u./gm) were all collected from the quarry face and exhibited a range of magnetic behaviour over the area of the present quarry outcrop. All samples showed large initial fall in moment after which intensities remained relatively steady although directions of magnetisation continued to change. Some were largely unstable to a.f.

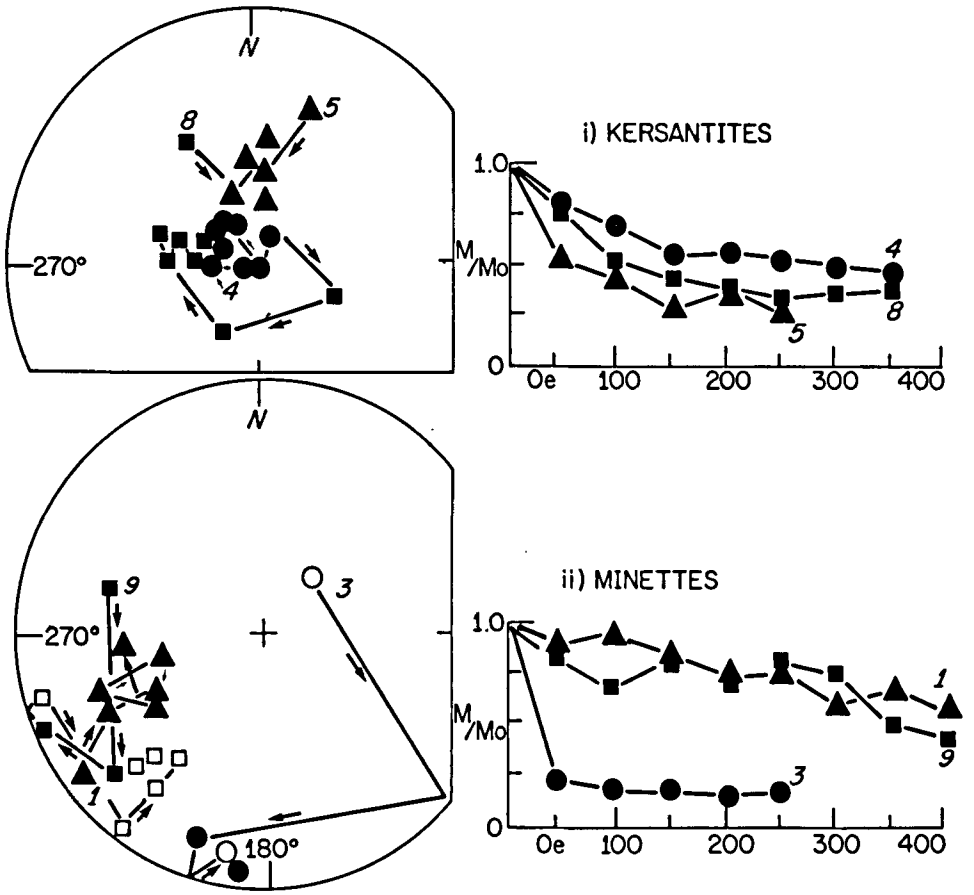


Fig. 5. Behaviour of directions of magnetisation and magnetic moments with progressive a.f. treatment for typical kersantite and minette lamprophyres of the Cross Fell Inlier. The symbols are the same as for Fig. 4 and the numbers refer to the palaeomagnetic sites of Fig. 3 and Table 2.

treatment while others, in particular the higher intensity samples, had to be demagnetised above 600 oe. or even 1000 oe. before directions began to stabilise with shallow southwesterly directions. It is clear from demagnetisation behaviour that magnetite is the principal remanence carrier (Nagata 1960). This mineral occurs as sparse subhedral or euhedral grains (15–200μ) with localised subdivision of grains by ilmenite lamellae equivalent to the oxidation class 2 of Ade-Hall *et al.* (1971). The ubiquitous haematite which gives the granite its red colour occurs at fine needles or blades (3–10μ) oriented along cleavage planes, abundant grains (mostly $>5\mu$) and as a cryptocrystalline red dust within orthoclase phenocrysts, but it does not appear to contribute significantly to the magnetisation. This is further emphasised by thermomagnetic determinations (Fig 6) which yield a Curie point at ~560°C but show no appreciable residual magnetisation between 560 and 670°C. The present sample is not regarded as large enough to define a precise palaeomagnetic pole position. The direction for 12 samples listed in Table 1 is derived from averaging the sample directions obtained after they had stabilised.

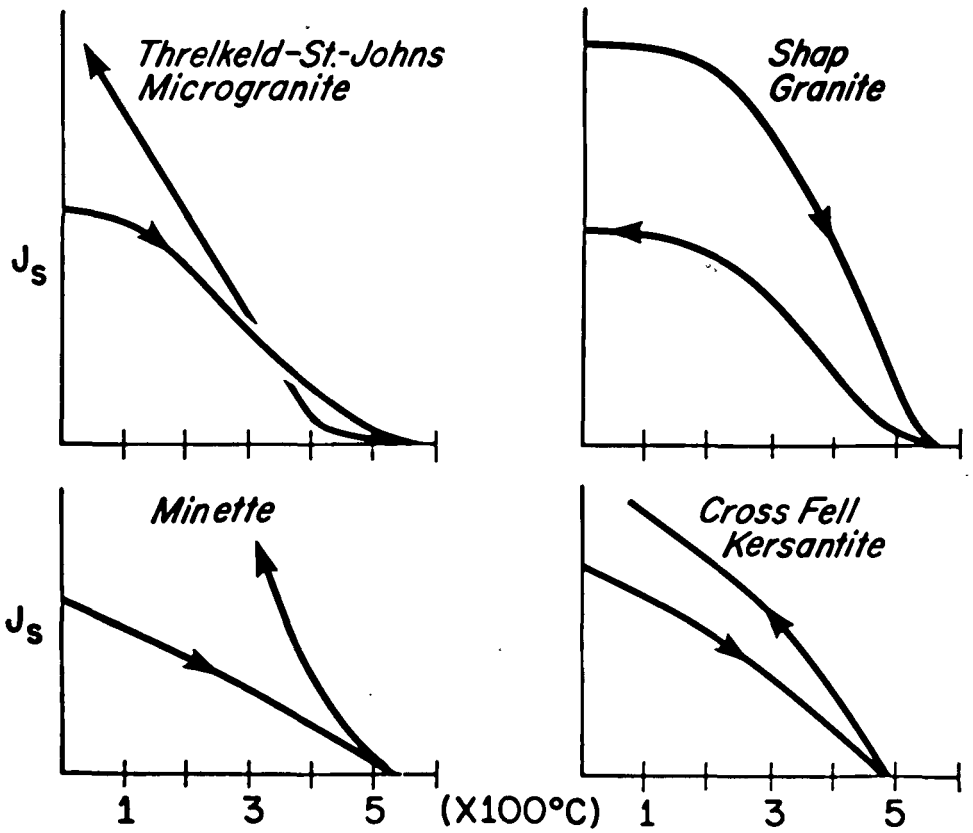


Fig. 6. Some thermomagnetic curves (temperature in degrees centigrade versus saturation magnetisation, Js) for rocks of this study. The ordinate axis is arbitrary.

In view of the weak NRM values quoted above it is somewhat surprising to find a strong magnetic anomaly (amplitude 550γ) situated over the Shap Granite (c.f. I.G.S. aeromagnetic map 8). The shape of the anomaly suggests a strong induced component of magnetisation and this is supported by susceptibility values determined on our cores which range from 1 to $1{\cdot}6 \times 10^{-4}$ cgs units. The calculated induced component based on these susceptibilities is still not sufficiently high to account for the observed anomaly and a detailed ground magnetic survey has resolved two anomalous regions located within contact metamorphosed and magnetite-bearing Borrowdale Volcanics (C. A. Lock and G. C. Brown, submitted to Nature). Thus the magnetic source is within the contact aureole and not the Shap granite as suggested by the low resolution aeromagnetic survey.

3g. Lamprophyres of the Cross Fell Inlier

The lamprophyres, whose petrology has been described in section 2.2, carry weak remanent intensities ranging from 1×10^{-5} to 1×10^{-7} e.m.u./gm. In general the kersantites are more stable to a.f. demagnetisation than the minettes and exhibit progressive fall in moments with little change in direction of remanence above 150 oe. (Fig. 5). Magnetic directions are then all steep and positive. The minettes illustrate a greater

Table 2. Palaeomagnetic Results, Cross Fell Inlier

	N	D	I	R	k	α_{95}	
E–W (Kersantite) Lamprophyre dykes							
1	4	318	75	3·60	7	36	
2	7	144	87	6·42	10	20	
3	5	296	68	4·82	22	17	
4	6	209	76	5·69	16	17	*Palaeomagnetic Pole* °E °N
5	6	124	84	5·60	12	21	337·0 53·6
Mean, 5 sites		274·7	84·9	4·88	34	13·3	($d\Psi = 25{\cdot}7$, $d\chi = 26{\cdot}1$)
WSE–ENE (Minette) Lamprophyre dykes							
6	9	225	35	8·48	15	14	
7	7	243	47	6·07	6	25	
8	4	204	6	3·85	21	21	
9	7	251	18	5·98	6	27	
10	6	232	30	5·61	13	19	
11	6	238	−11	5·38	8	25	
12	7	171	2	6·60	15	16	
13	7	203	−24	6·44	11	19	*Palaeomagnetic Pole* °E °N
14	5	199	−24	4·76	16	19	316·3 −22·9
Mean, 9 sites		217·5	9·7	7·49	5	24·7	($d\Psi = 12{\cdot}6$, $d\chi = 24{\cdot}9$)

Footnote: D and I are the mean declination and inclination of the remanence vector derived from N sites or samples with a resultant vector R and precision k defined by $k = (N-1)/(N-R)$. α_{95} is the radius in degrees of the 95 per cent confidence circle about the mean direction and $d\Psi$ and $d\chi$ are the semi-axes of the 95 per cent confidence oval about the palaeomagnetic pole in the co-latitude direction and at right angles to it respectively.

Table 3. Potash: Soda ratio in Lake District and Cross Fell Intrusives

Intrusion	*No. of analysed samples*	*Ratio* $\frac{Ave\ K_2O}{Ave\ Na_2O}$
Ennerdale Granophyre	4	0·43
Cross Fell Kersantite	1	0·54
Haweswater Dolerite	2	0·58
Dufton Microgranite	1	0·77
Threlkeld–St John's Microgranite	4	0·81
Skiddaw Granite	2	1·28
Shap Granite	3	1·38
Eskdale Granite	8	1·46
Weardale Granite	19	1·48
Cross Fell minette	1	3·63

All analysis by X-ray fluoresence method described in Brown *et al.* 1973. Full analyses by G. C. Brown, D. J. Hughes and J. G. Fitton (unpublished).

range of magnetic coercivities than the kersantites often with large initial change in magnitude and direction of the vector following more stable behaviour; these samples are most stable over the range 250–400 oe. and directions have accordingly been averaged over this range and combined to yield the group mean directions listed in Table 2.

The rocks distinguished as minettes from petrographic examination (section 2b) all have distinct directions of magnetisation (shallow south westerly) from the kersantites (steep positive), and only minette site 7 has a circle of 95 per cent. confidence about the mean direction overlapping with circles of confidence of the kersantites (Table 2). Clearly the two suites are palaeomagnetically different, reflecting a time difference between their respective emplacement—a feature we examine in more detail shortly.

4. Discussion

Several results from this study merit a straightforward interpretation in the context of other palaeomagnetic work on British Palaeozoic rocks (summarised by Briden *et al.* 1973). Firstly, results from the Borrowdale Volcanic Series are similar to the data of Faller *et al.* (1976) from the same region. The combined mean direction yields a palaeomagnetic pole assigned to the Late Llanvirn–early Caradocian interval and corresponds closely with Lower Ordovician poles from Britain (Fig. 8A). It is not certain whether intrusion of the Haweswater dolerite preceeded the pre-middle Caradocian folding (Soper and Numan 1974) of the Borrowdale Volcanic Series into

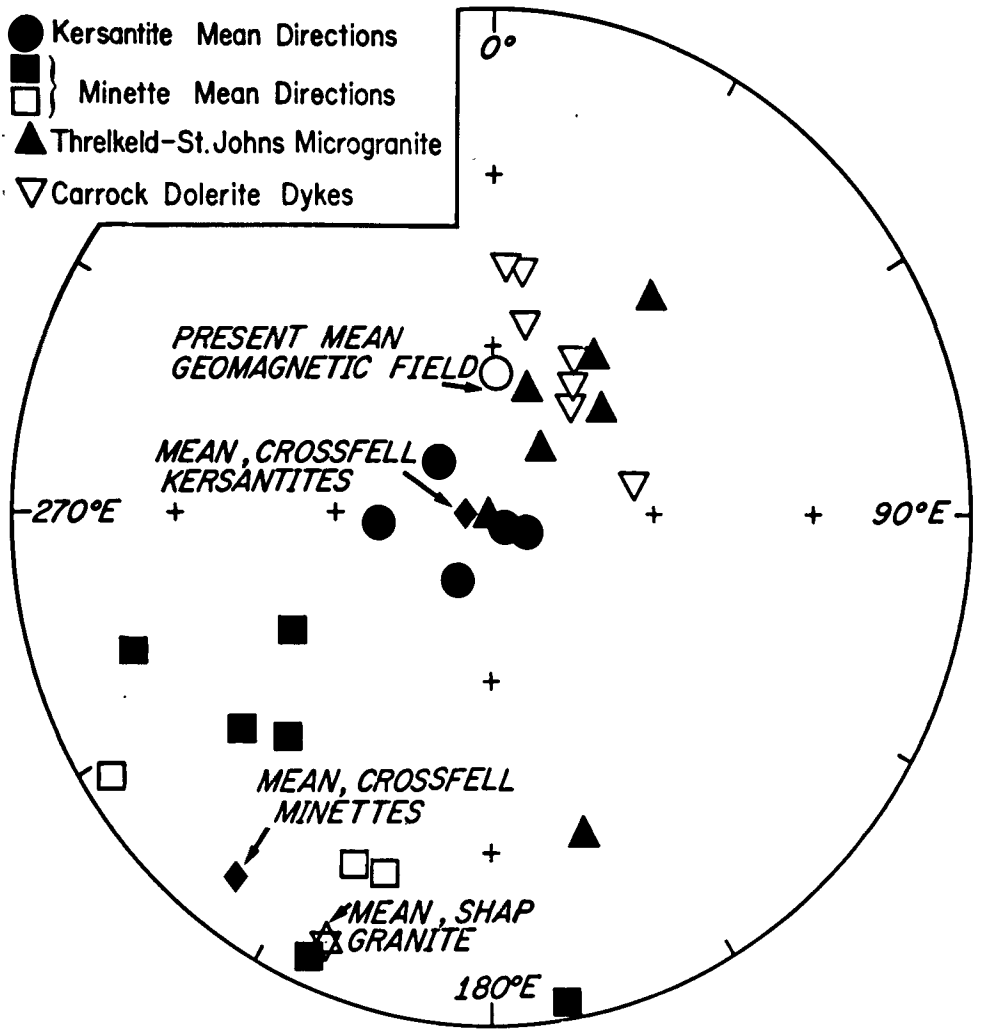


Fig. 7. Stereographic projection showing site mean directions of magnetisation derived from dolerite dykes of the Carrock Fell area, the Threlkeld-St John's microgranite, kersantite and minette lamprophyres of the Cross Fell Inlier, and the Shap Granite.

E–NE trending open folds, or, if this was the case, whether folding of the dolerite was coupled with that of the lavas. The body lies along the north flank of the Wrynose–Nan Bield anticline and correction for dips measured on intruded tuffs would move the palaeomagnetic pole to 177·1°E, 37·9°S (D = 0·3°, I = −81·5°). Prior to dip correction, the pole correlates with other Ordovician poles (Fig. 8A) suggesting a post-folding intrusive relationship. Pending further results, the single site (22) from the Embleton diorites is regarded as also indicating an Ordovician age (Fig. 8A).

Interpretation of palaeomagnetic results from the Carrock dykes, Threlkeld–St John's microgranodiorite and Cross Fell kersantite lamprophyres may relate to a recent study of British Palaeozoic rocks yielding steep, and apparently anomalous, directions of magnetisation. Thomas and Briden (1976) found steep negative magnetisations in nine Welsh major granodiorite and microgranite intrusions, and the Cader Idris basalts

of the North Wales Igneous Province; they suggested that the results implied either unrecognised tectonic rotations or a palaeomagnetic field anomaly. A Caradocian age for this event is defined by the age of the Cader Idris basalts although it may have been a relatively short-lived event because other Caradocian volcanics from the Welsh Borderlands (Piper 1978) have shallow magnetisations yielding typical Ordovician palaeomagnetic poles close to those illustrated in Figure 8A. A range of dolerites from the Welsh Borderlands intruded between Upper Ashgill and Upper Llandovery times also yield anomalous directions of magnetisation of both polarities (Piper 1978) with a mean pole near 60°E, 15°N, and show that the Lower Palaeozoic a.p.w. path for Britain is more complicated than originally suspected (Briden *et al.* 1973).

The new results from the Threlkeld–St Johns microgranodiorite and Carrock Fell dykes extend the distribution of Upper Ordovician palaeomagnetic studies yielding anomalous directions, and the palaeomagnetic poles are plotted in Figure 8B together with the poles of Thomas and Briden (1976) from North Wales. It is observed that the Lake District results are similar to some of the North Wales results but are of opposite polarity; this lends support to the postulate that these events are approximately Caradocian in age; the palaeomagnetic poles are not comparable to the late Ashgill–Llandovery poles. Wadge *et al.* (1976) regard the Thelkeld–St Johns microgranodiorite as early Caradocian by correlating their 445±15 m.y. age with the 447 m.y. date for the base of the Caradocian (Adams *et al.* 1960; Lambert 1971) and they discuss the possibility that intrusion of the microgranodiorite may have been contemporary with extrusion of the later part of the Borrowdale Volcanic Series. However, palaeomagnetic study of the uppermost part of the Borrowdale Volcanic Series in the Kentmere area (J. D. A. Piper, in preparation) has produced comparable directions of magnetisation to those given here and by Faller *et al.* (1976) for the lower part of the succession. We infer, therefore, that the intrusion of the Threlkeld–St John's body post-dates the Borrowdale volcanism. Ross *et al.* (1976) have dated the type Actonian of the Welsh Borderlands at 451±21 m.y. and the oldest sediments of the Coniston Limestone Group are the Stile End Beds of Actonian age. Hence, on the basis of the radiometric evidence it is possible that intrusion of the Threlkeld–St John's microgranodiorite was broadly contemporaneous with residual post-Borrowdale Volcanic Group volcanism including the Stockdale Rhyolite. This latter unit has yielded a direction of magnetisation D = 170°, I = −73° after a.f. cleaning and tilt (correction which is close to the anomalous directions derived by Thomas and Briden (1976) from North Wales in rocks also of probable Caradocian age. Although the directions from

Fig. 8. Palaeomagnetic poles from this study plotted on equal area maps with (where appropriate) 95 per cent. confidence ovals, together with some other palaeomagnetic poles assigned to comparable intervals of Lower Palaeozoic time. In Fig. 8A the lettered poles are from A, Ashgill intrusions, Builth inlier; C, Borrowdale Volcanic Group, western sites; E, Eycott Group Lavas; F, Younger Gabbros, Aberdeenshire; G, Byne Hill gabbro; H, Builth Volcanic Series; I, Carrock Fell Complex. In Fig. 8B poles derived from steep directions of magnetisation of this study are compared with anomalous palaeomagnetic poles of probable Upper Ordovician age from the North Wales Igneous Province. Numbered poles are: 1, Penmaenmawr granodiorite; 2, Penmaenbach rhyolite; 3, Foel Fras granodiorite; 4, Garnfor granodiorite; 5, Carreg-y-Llam granodiorite; 6, Bodeilias granodiorite; 7, Mynydd Nefyn granodiorite; 8, Gurn Ddu granodiorite; 10, Cader Idris basalts. In Fig. 8C palaeomagnetic poles from the Cross Fell minettes and the Shap Granite are compared with poles of Upper Silurian to Upper Devonian age lettered: A, Lower O.R.S. Lavas, Cheviot; B, Wenlock Lavas, Somerset and Gloucestershire; C, Garabal Hill–Glen Fyne; D, Lower O.R.S. Lavas, Glencoe; E, Lorne Plateau Lavas; F, Lower O.R.S. Lavas, Midland Valley; G. Arrochar Complex; P, Upper O.R.S. Jedburgh; R, Upper O.R.S., Bristol District; S, Middle O.R.S. Foyers; T, Middle–Upper O.R.S. Lavas, Orkney. The palaeomagnetic poles are summarised in Briden *et al.* (1973) and Thomas and Briden (1976). Confidence ovals for poles from this study are stippled in Fig. 8B.

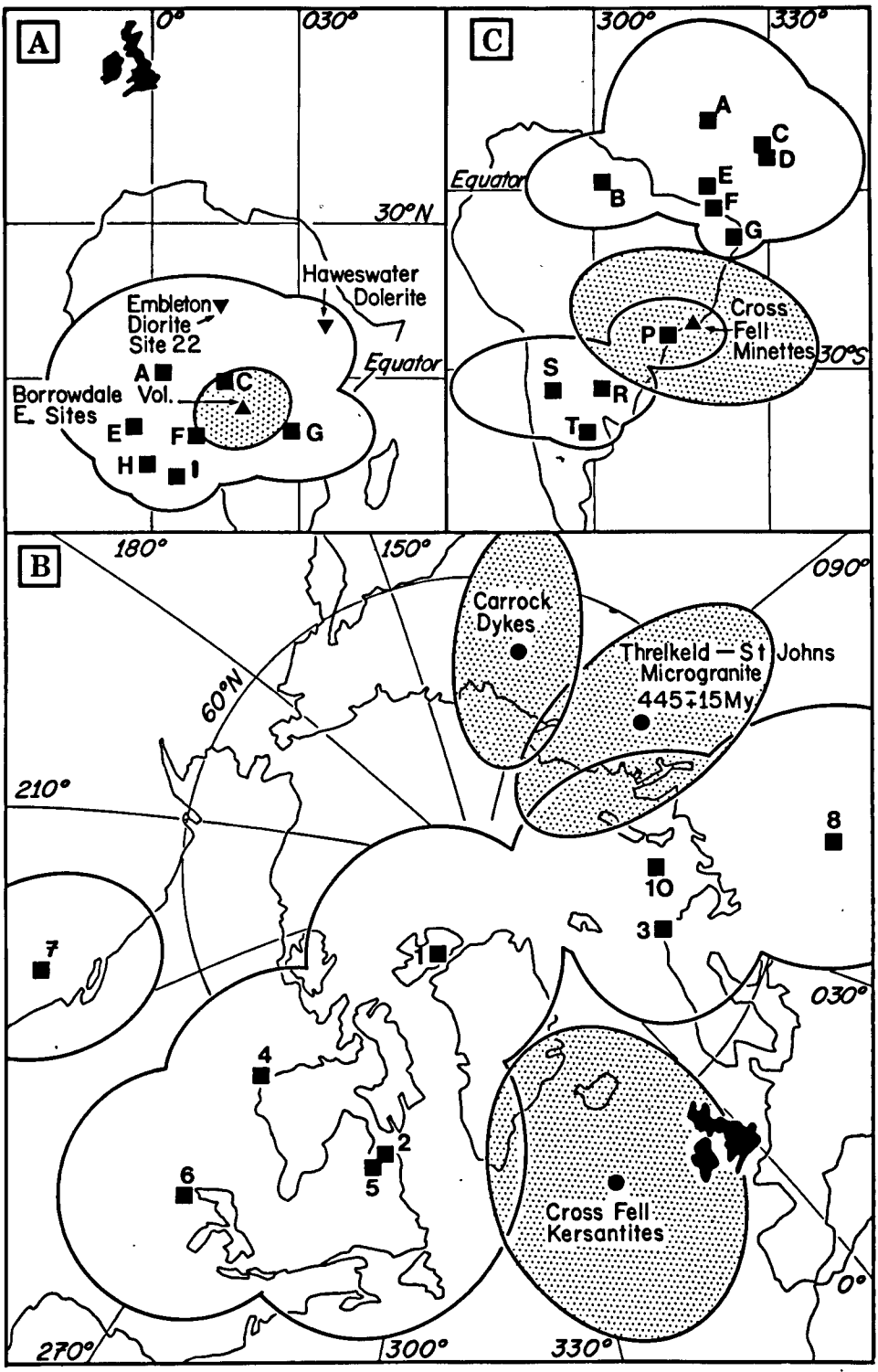

A
0°
030°
30°N
Haweswater
Dolerite
Embleton
Diorite
Site 22
Equator
Borrowdale Vol.
E. Sites
A
C
E
F
G
H
I
C
300°
330°
Equator
A
C
D
E
B
F
G
Cross
Fell
Minettes
P
30°S
S
R
T
B
180°
150°
090°
Carrock
Dykes
Threlkeld — St Johns
Microgranite
445±15My
60°N
210°
8
10
3
7
1
030°
4
2
5
6
Cross Fell
Kersantites
0°
270°
300°
330°

Fig. 8

the Threlkeld–St John's intrusion, Carrock dykes, and the Cross Fell kersantites are of opposite polarity to these data, they are also anomalous and may well be part of the same geomagnetic event. We also note that the dip-corrected pole from the Haweswater dolerite (implying a pre-middle Caradocian age) also lies in the same field as these anomalous poles (Fig. 8B), and in accordance with the chemical affinities with the Cross Fell kersantites, this suggests a closely similar time of emplacement.

The E–W dolerite dykes intruding the Eycott Group near Carrock Fell have directions of magnetisation which are mostly slightly shallower than those from the Threlkeld–St John's microgranodiorite (Fig. 7); this is reflected in the slightly different pole position (Fig. 8B) but since the 95 per cent. confidence circles overlap and they have the same polarity, it is inferred that they are of similar age. Potassium–argon whole rock ages of 350 ± 5 m.y. and 290 ± 5 m.y. were obtained from two of the dykes of the palaeomagnetic collection (S. R. Charlton, personal communication) but since the palaeomagnetic directions are quite different from any Carboniferous or Devonian directions from Britain we infer that both these sites have experienced argon loss.

Harker (1892) originally proposed that all the dykes of northwest England were part of the last intrusive phase of the Shap Granite because the lamprophyre dykes are distributed within a circle approximately 70 km around Shap, and the few he examined contained biotite and pink feldspar. However, it was noted in section 2b. that the kersantite and minette dyke trends are distinct, the former having an E–W strike while the latter are disposed ENE–WNW and thus point towards the Shap Granite. Hudson (1937) suggested that the dykes of the southern part of the Cross Fell Inlier were related to the Shap Granite and the dykes in the northern part of the inlier were related to the more northerly Ennerdale Granophyre and Threlkeld–St John's microgranodiorite (Fig. 2). The present study confirms, in part, the suggestions of Hudson. Thus, although the kersantites cannot be dated precisely from their pole position (Fig. 7 and 8B). we assign them to the interval of anomalous palaeomagnetic field behaviour occupying part of Caradocian times; they are thus of comparable age to the Threlkeld–St John's microgranodiorite. There is, in turn, close correspondence between the palaeomagnetic directions of the Cross Fell minette lamprophyres and the Shap Granite (Tables 1 and 2, Fig. 7) and the pole positions lie in between poles derived from other British Lower Devonian and Middle–Upper Devonian rocks (Fig. 8C).

Clearly, the two major episodes of Caledonian granite intrusion in northern England find their expression in episodes of dyke intrusion. The palaeomagnetic distinction between Late Ordovician and Devonian activity is paralleled by petrochemical variation reflecting magmatic evolution and summarised by the K_2O/Na_2O ratios presented in Table 3. The early kersantite dyke analysis from Cross Fell has a low ratio lying between the Ennerdale and Threlkeld–St John's intrusion (c.f. Hudson (1937) noted above). Similar ratios have been determined for the camptonitic Haweswater intrusion and the Dufton microgranite of Cross Fell which is provisionally allocated to this Ordovician intrusive suite; the latter could be the acid differentiate of the Kersantite magma (Hudson 1937). The Carrock Fell dykes are added to this suite on the strength of their palaeomagnetic expression. We deduce that pre-mid Caradocian folding preceeded these intrusives but that the high strain end-Silurian folding separates them from the later Devonian episode. The Shap granite and Cross Fell minettes are joined by the Skiddaw, Eskdale and sub-Pennine Weardale granites as having distinctly higher K_2O/Na_2O ratios (Table 3) and these relatively plutonic intrusions all postdate the Silurian cleavage (N. J. Soper in discussion of Wadge *et al.* 1976). Emanations from the batholith ridge connecting Shap to the Alston area were probably responsible for the highly potassic hydrous minette magmas.

These considerations define two points in a trend of K-enrichment from Caradocian intrusive magmas of $K_2O/Na_2O < 1$ to Devonian rocks of $K_2>/Na_2O > 1$ which parallels on earlier evolution in extrusive volcanism from the basalt-andesite (Lower Llanvirn) Eycott group to the andesite-rhyolite Llandeilo–Caradoc Borrowdale Volcanic Series and rhyolitic post-Borrowdale volcanism. The question of magma source is still somewhat open but the initial strontium isotope ratios of the Threlkeld–St John's intrusion (0·7055±0·0017, Wadge *et al.* 1976) is rather similar to that of the later Weardale intrusion (0·706, no error quoted; Holland and Lambert 1970). Both signify almost direct evolution from a mantle source region, a feature common to both modern Cordilleran granites (Brown 1977) and the younger British Caledonian intrusions in particular (Brown and Hennessy 1978). K-enrichment trends are also observed in recent destructive margin volcanic and plutonic rocks: such trends are usually taken to imply a deepening magma source region. The conclusion from the palaeomagnetic dating of Lake District intrusives is that two distinct periods of magmatism identify magmas of well-separated K_2O/Na_2O ratio and thereby lend support to models which require generation of Lake District magmas from the mantle in the region of a descending lithospheric plate (e.g. Fitton and Hughes 1970).

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